Khalani:
A Next-Generation Blockchain with On-Chain Proofs

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Abstract

Khalani is an attempt to design and prototype a next-generation blockchain architecture that has the potential to eventually replace the currently dominant Ethereum-style chains. We postulate a collection of novel features for this new blockchain:

- purely functional programming on-chain
- linear types (i.e. language-level resources)
- formal logic on-chain (i.e. the ability to formulate and prove theorems and bind them with on-chain programs)
- on-chain pub-sub and timers (i.e. a proper event-based programming)
- support for intent -based programming
- code-as-asset (so that a code deployed on-chain generates financial profits and can be traded)
- account abstraction (ability to programmatically define the behavior of blockchain accounts)

We present the details of how these features fit together and how they provide remedies for known problems of existing blockchains. Although some of them are already present elsewhere, the overall vision is new. We conclude with an overview of Axi, Khalani's on-chain programming language.

Caution: this *litepaper* sketches only the key ideas from the on-going research. It is different from the upcoming Khalani *whitepaper*, which will contain detailed technical specifications.

1 Introduction

1.1 Timeline of blockchain technology

Blockchains have been around since 2009 (when Bitcoin was launched). Key milestones include:

- 2013 - understanding that decentralized money could be generalized to a decentralized computer

- 2015 - launch of Ethereum, the first widely adopted decentralized-computer platform
- 2020 - the rise of proof-of-stake consensus protocols, promising lower energy use
- 2022-2023 - wider adoption of zk-rollups (e.g., mainnet launches of zkSync Era, Polygon zkEVM), based on the idea of *zero-knowledge proofs*

As of 2025, the EVM camp - consisting of several blockchains compatible with the Ethereum Virtual Machine - is still dominant as measured by TVL (Total Value Locked).

1.2 Major obstacles for blockchain adoption

The adoption of blockchains remains limited. There are several obstacles hindering wider use.

1.2.1 Bugs are deadly yet hard to prevent

Bugs in on-chain programs are extremely costly - a single smart-contract bug, once exploited by a hacker, can easily destroy the on-chain business that depends on it. However, current methods for preventing bugs are insufficient - testing cannot guarantee the absence of bugs, and neither can audit. Moreover, once a bug is found, it cannot be fixed, because on-chain programs are immutable by design. Even if a program is explicitly designed to support an upgrade operation (for example using proxies), this upgrade usually comes too late to prevent losses caused by hackers.

Some examples:

- June 2016 - The DAO exploit ($50M lost), resulting in the subsequent Ethereum hard fork
- November 2017 - Parity wallets failure ($150M lost)
- August 2022 - Nomad bridge failure ($190M lost)

1.2.2 Scammers everywhere

Because anyone can deploy a program to a public blockchain, users face a core question: which services are trustworthy, i.e. will they actually do what they claim? How can users be confident a service will not steal their funds?

Some examples:

- December 2020 - Compounder Finance ($10M lost)
- October 2021 - AnubisDAO ($60M lost)
- November 2021 - Squid Coin ($3M lost)

1.2.3 "Lost in the maze" syndrome

Blockchain technology evolves rapidly. For typical users the space is a maze of chains, services, tokens, credentials, keys, exchanges and responsibilities which becomes increasingly hard to navigate. Although there are numerous efforts to establish community standards (for example ERC-20 tokens, wallets, DEXes), they do not offset the quickly growing complexity.

1.2.4 Cross-chain integration

According to CoinMarketCap, as of October 2025 there are 70 blockchains with market cap exceeding $1B. The ecosystem does not seem to converge to a single, universal public blockchain and the users have already got used to this diversity of independent chains. This creates integration problems, the first and foremost of which are cross-chain swaps, i.e. moving assets across blockchains.

1.2.5 The programming model is limiting

For software developers, blockchains are not nearly as convenient as more traditional computing platforms. The example limitations are:

The paid-call problem. to use a given blockchain, a users needs to acquire the native cryptocurrency of this blockchain (in order to pay gas fee). This means the gas mechanics constrains the architecture of dapps. This problem could be mitigated by features like sponsored transactions or delegation.

Explicit transaction signing. Every action taken by a user necessarily takes of form of a transaction, which in turn requires cryptographic signing. Although signing could be theoretically done by any software whenever needed, this would create a security vulnerability around private keys sharing. Therefore users tend to use a single "trusted" wallet app for all the signing work, which mitigates sharing risk but on the other hand enforces every act of signing to be a tedious manual interactive process, which in result impairs overall user experience. This could be mitigated if the blockchain programming model supported delegation.

Lack of a uniform delegation mechanism. The previous case is just part of a bigger one: delegation is just not there in blockchains. Say we have Alice (user), Bob, Charlie and Dylan (on-chain services). Bob provides some complicated functionality which involves further interaction with Charlie and Dylan. Alice has no rights to interact with Dylan, so she is forced to use Bob as a privileged intermediary. Charlie recognizes Alice as a user with certain access rights. Bob cannot directly call Charlie because Bob has no rights to do so. Hence Bob would like to interact with Charlie on behalf of Alice and this is exactly the essence of delegation. When Alice wants to ask Bob to do something for her, Bob will have troubles providing satisfactory level of "automation" of actions on behalf of Alice. This is not because such a delegation is fundamentally impossible, rather just the blockchain industry has not agreed on sufficiently safe and universal mechanism of delegation yet. For example, Ethereum proposals ERC-4337 and EIP-7702 suggest such a universal delegation mechanism that could be added to Ethereum.

The "self-driving taxi paradox". Given the currently available autonomous driving technology, combined with wireless Internet access, it is possible to run a taxi service without a driver. However such a taxi business would face a seemingly trivial problem - how a car without a driver would accomplish filling itself up with gas? So here is the paradox - a self-driving taxi still needs an operator just to accomplish the gas station interaction. In blockchains we have a similar trouble: services cannot pay for their own gas consumption. Normally, it the transaction sender who pays gas for the whole transaction execution. This immediately becomes a problem when a service has a common part (say "a backend") that from time to time needs to perform some maintenance tasks. These tasks are not performed for a specific user of the service, these tasks are for the service as a whole. But because the service has no way to cover the gas cost of these maintenance activities, the developer of the service has only two solutions available:

1. charge random users during their interaction
2. invoke maintenance whenever needed using a dedicated "housekeeper "account.

Unfortunately 1. is not reliable (this approach can blow a transaction because of exceeded gas limit) and not fair, while 2. is even more unreliable because it introduces a dependency on a centralized service. The whole issue could be mitigated by supporting the *gas reserve* mechanism.

No timers and pub-sub Smart contracts cannot natively react to time or on-chain events. While reactive features (in particular the observer pattern) are standard in off-chain programming, they are missing from on-chain programming. Because of this, developers have to route reactive processing via off-chain observers, which weakens decentralization.

Limited privacy Everything on the blockchain is anonymous but public, which is not at all the same thing as privacy. This significantly limits the development of services that critically depend on the privacy of customers' actions.

1.2.6 The open source paradox

Software distribution largely falls into two camps:

- **Commercial software** must be purchased, which funds its creation and maintenance.
- **Open-source software** is free. It is created and maintained by enthusiasts in their spare time.

The open source paradox is this: open source creators, despite providing crucial value, are not paid for their work. An additional layer of the paradox is that a lot of commercial software depends on open-source software.

In the blockchain world the situation is even more paradoxical, as there are two dimensions of software to consider:

- node software that runs the P2P network - smart contracts that run on-chain

In theory node software can be closed source, but in practice this never happens because transparency is valued so much that only open-source really qualifies for the purpose.

As for on-chain programs, in theory they are not open-source, although they can be easily decompiled and copied because they are visible in the blockchain state. In practice, trust concerns and implied transparency requirements mean that users prefer smart contracts with published source, regardless of decompilation. Put simply: users won't trust a dapp that does not publish its smart contract source code.

In summary, blockchains lean very strongly towards open-source; monetization is not based on selling or hiding code. Founders most often try to monetize their blockchain via native-asset economics, for example through an Initial Coin Offering. But with more adoption, the blockchain starts running on its own, which means that core on-chain components are developed and maintained by a broader open-source community.

Hence the paradox: the builders who add the most value are not the ones rewarded. On-chain code amplifies this, as it is effectively public by default.

1.2.7 Integration paths are not well-established (yet)

Public blockchains are available to anyone, but integrating a blockchain into a business is difficult in practice (except for simple payments), as it is often necessary to hire developers who will design, implement, and deploy a custom dapp. Easier integration, without having to build a full dapp, requires the blockchain community to converge on convenient and widely adopted abstractions. An example of such an abstraction is intent-based processing (which we discuss later).

1.2.8 Performance

Blockchain performance is low. Throughput (measured in transactions-per-second) is constrained by networking and execution. Latency (the delay between transaction sending and its confirmation/finalization) is driven by consensus design and network limits.

1.3 Our mission

We aim to prototype a next-generation blockchain. Our focus is on the programming model, i.e. the Virtual Machine (VM). We pair a novel and innovative VM design with some existing performant proof-of-stake consensus protocol.

We hope that the adoption of Khalani will challenge EVM dominance by offering significant usability gains and that, as a consequence, its ideas will spread to other chains, inspiring innovation and moving the industry forward.

The key distinguishing features of the Khalani blockchain are:

- Axi, a purely functional programming language for smart contracts
- – type system based on linear logic/quantitative type theory (aka "programmable assets")

  - on-chain formal logic for proving theorems about smart contracts
  - concepts (infrastructure that combines interfaces with formal specifications, in other words combines programming and logic into a convenient flow of "semantic interfaces")

- native support for intents and delegation
- on-chain events system (pub-sub and timers)
- gas reserve mechanism
- native account abstraction
- code-as-asset (code you wrote earns money for you, code is an asset and can be traded)

The following Table 1 gives a bird's eye overview of key blockchain adoption obstacles (as listed above) vs features of Khalani blockchain that are supposed to mitigate them.

1.4 Sources of inspiration

The Sui blockchain was the main inspiration behind our ledger structure and the use of linear types to guide the programming language's interaction with the ledger. We also extensively studied UTXO-based blockchains (chiefly Nervos and Cardano) to understand their advantages and limitations.

When designing Axi, our guiding principle was Intuitionistic Type Theory. The logical features were inspired by proof assistants such as Coq and Lean, whereas the programming side drew inspiration from languages like Haskell (the general spirit) and Unison (regarding optimal blockchain-aware source code representation). Earlier we rejected an approach based on Denotational Proof Languages and the Athena proof system.

We also took some time diving into the current state of intent-based processing ideas:

- `https://eips.ethereum.org/EIPS/eip-7521`
- `https://anoma.net/`
- `https://docs.essential.builders/welcome/`

2 Ingredients of the architecture

2.1 Axi

2.1.1 Functional programming with formal logic built-in

In **imperative programming**, the most widespread programming paradigm, the role of the program is to modify the computer's memory. The program is composed of a bunch of statements. The most basic statement is assignment, whose role is to set a variable to a particular value and a variable is a name that refers to a particular location in memory. These basic assignments are combined into more complex programs using other statements, like loops and

Table 1: Problems and proposed solutions

Problem	**Khalani's solution**
Bugs (exploited by hackers)	Khalani smart contracts are written in Axi, a purely functional programming language. Axi supports concepts and using formal logic to prove that a component is compliant with a concept. The existence of concepts will generate social pressure on developers, as the reputation of their dapps will depend on the compliance with well established concepts. The source of the pressure will come from users because the concepts infrastructure will allow easy to use guidance for average users (similar to how easily people use SSL in internet browser).
Scams and frauds	Code in a purely functional programming language is very transparent, making it harder to obfuscate what a smart contract actually does. Formal logic provides an effective way to express formal guarantees. Checked proofs are part of the blockchain state agreed upon by consensus. This means that every user could perform a basic smart contract audit by checking whether it implements a few traits that are commonly understood as providing some anti-fraud protection.
Lost in the maze syndrome	Intents
Cross-chain integration	Intents
Delegation mechanism is missing	Delegation mechanism
Self-driving taxi paradox	Gas reserve
No on-chain timers	In-protocol support for timers
No on-chain pub-sub	In-protocol support for pub-sub
Open source paradox	Code-as-asset
Integration paths not well established	Economic incentives to build and converge around common abstractions (concepts), implemented in Axi as concepts.
Performance	UTXO consensus, which is based on partial order (as opposed to a sequence), with declared inputs of every transaction. Also, the object ownership model reduces the collisions between transactions.
Limited privacy	For now the relevant features of Khalani are not decided yet. We are looking into existing solutions (such as Monero and ZCash) to estimate feasibility.

branching. Execution of the program proceeds statement by statement, at every step modifying tiny chunks of memory. When the execution is complete, the transformation of the computer's memory described by the program has been carried out.

Functional programming is a programming paradigm in which the role of the program is to compute a single output value. The entire program is an expression which, in turn, is composed of a bunch of subexpressions. The two most basic kinds of expressions construct anonymous functions and apply them to arguments. Variables are names that refer to a particular expression and they are immutable. Simple expressions are combined into more complex ones using yet more expressions, like pairing or pattern matching. Execution of the program proceeds by transforming the program according to a set of rewrite rules that dictate how the various kinds of expressions interact with each other. When the execution is complete, all that remains of the program is a single value which is the program's result.

Functional programming is less familiar to most programmers than imperative programming (which subsumes object-oriented programming), but has one significant advantage: it is much easier to reason about functional programs. In particular, it is much easier to prove theorems about such programs, including formally proving that they are correct according to some specification.

We adopt purely functional programming in Khalani precisely for this reason - because we believe this is the ultimate way to ensure correctness of on-chain programs. Axi, our programming language, was designed from scratch to precisely fit in with Khalani's ledger model and its various novel features.

Axi borrows key ideas from type-theoretical proof assistants like Coq, Lean and Agda. These languages are purely functional (like Haskell) but with an added superpower - a developer can also represent formal mathematical theories with them. So in Axi, one can implement programs and also prove theorems about these programs.

2.1.2 Concepts

Axi merges programming and logic via **concepts** ("lawful traits"). A concept bundles together declarations of constants, functions, abstract types, and logical statements about them. In other words concepts is how the profits of formal logic are transported to the programming layer.

Concepts can capture high-level on-chain concepts such as DEX, bridge, swap, and escrow. We expect a growing library of such concepts to develop organically over time, thus encouraging convergence on common abstractions needed for integration.

2.1.3 Linear types

Linear types enforce that a value is used exactly once, thus allowing the type system to precisely track resources. This lets developers:

- safely deallocate an object after its use
- design software interfaces that guarantee a resource cannot be used after it has been closed or transitioned to a different state

While brewing since the '80s, linear types have been popularized around 2012 when they appeared as the core concept in the Rust programming language. In 2018 this idea was then adopted by the Move programming language, where linear types are the way to keep track of **resources** during program execution. This neatly fits into the on-chain money abstraction.

Move underpins Sui and Aptos, demonstrating the power of linear typing for building blockchain abstractions. This is why we decided to adopt linear typing in Axi.

2.2 Blockchain model

2.2.1 UTXO and state evolution

At the general level, Khalani blockchain is based on the UTXO model. This means that global history is composed of cells and transactions:

- cells represent pieces of state
- transactions represent units of computation

The model can be sketched with the following pseudocode:

```
struct Cell {
  address: Address,
  datatype: Datatype,
  version: Long,
  data: Bytes,
  status: CellStatus
}

struct Datatype {
  package: Package,
  module: String,
  structure: String
}

enum CellStatus {
  Alive,
  Dead
}

struct Transaction {
  nonce: Nonce,
  inputs: Vec<(Address, InputMode)>,
  outputs: Vec<Address>,
  txid: Hash
}

enum InputMode {
  Dependency,
  Update,
  Consumption
}
```

The blockchain state evolves by committing transactions. Each transaction consumes input cells and creates output cells. Cell identity is retained by its address. Each address has a linear version history. `Dependency` is a special read-only input mode in which the cell is not killed.

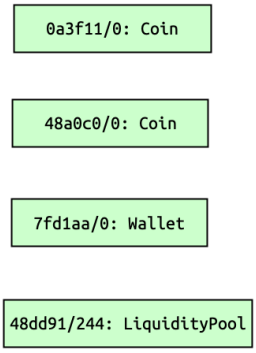
Cells read with input mode `Consumption` switch status to `Dead`. Cells read with input mode `Update` also switch status to `Dead`, and a modified copy (same address and type, incremented version) is created among the outputs.

Every cell is typed (with an Axi type) and contains exactly one value of that type. The type is set when the cell is created and cannot change during its lifetime. In other words, updates preserve the cell's type.

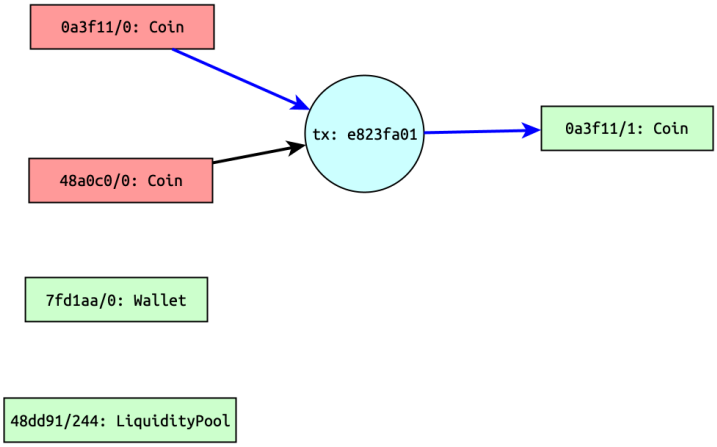
To illustrate the intuition behind the model, in the pictures we use the following conventions:

- 3-byte hashes as cell addresses - 4-byte hashes as transaction identifiers - `0a3f11/1` denotes version `1` of the cell at address `0a3f11` - cell types are shown as simple names - alive cells are green - dead cells are red - solid black lines represent cell consumption/creation - solid blue lines represent cell update - dotted lines represent dependencies (i.e. `input-mode=Dependency`)

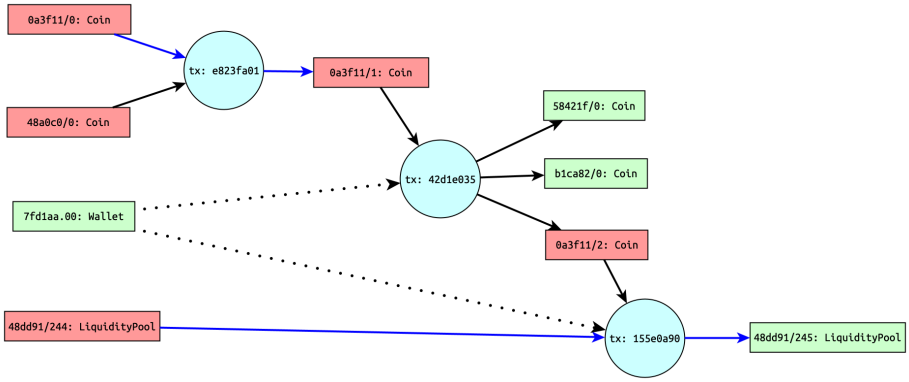
Suppose we start with four live cells:



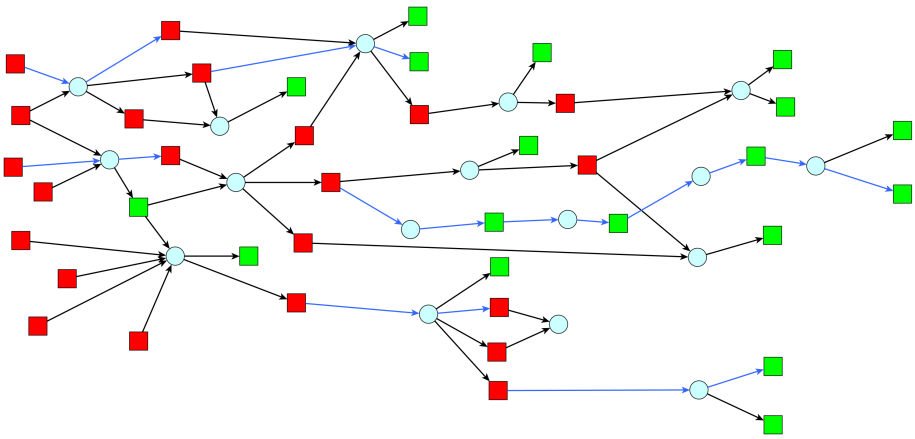
Then transaction `e823fa01` is executed. It has two inputs and one output. Cell `0a3f11` gets updated (so a new version of it is created). Cell `48a0c0` gets consumed (i.e. destroyed):



Then transaction `155e0a90` is executed. It has two inputs and one output:



The ledger grows indefinitely, forming a Directed Acyclic Graph (DAG) which might look like this:



The set of live cells is the current global state. It represents the latest state of the blockchain's memory.

2.2.2 Packages, modules and source code

On-chain code is organized into packages and modules. Every code deployment creates a new package. A package contains a hierarchy of modules (identified by their hierarchical names) that serve as namespaces and contain Axi code: types, constants, functions, traits, theorems and proofs.

The source code of a package is stored on-chain as an abstract syntax tree (AST) whose nodes are content-addressed by cryptographic hashes, so duplication is impossible. The package itself is addressed by the hash of its contents.

Every package is public and an import system allows code from any package to use code from any previously deployed packages. In other words, a package is a public shared library.

Packages are immutable and permanent, i.e. once deployed, they cannot be changed in any way and they stay on-chain forever. Thus there is no package

"upgrade" mechanism and no package versioning. Deploying a modified version of an existing package yields a new independent package with a new hash, regardless of name.

2.2.3 Accounts and ownership

As in other blockchains, Khalani has the concept of user accounts. An account is established as a public-private key pair, obtained within a selected cryptosystem. Khalani supports a pluggable architecture of cryptosystems in use.

Account address is obtained by SHA-3-256 hash of the concatenation of two values: cryptosystem id (1 byte) and the public key from the key-pair generated by the user.

Every transaction must be signed by an account. The address of signing account is call "tx sender".

Changes done by a transaction to the blockchain state are limited by access control mechanics. Every cell is either shared or address-owned. In the address-owned case, the owner may be an account or another cell.

2.2.4 Where are the smart contracts?

The reader may be surprised we have not mentioned smart contracts yet, while they otherwise seem to be an ubiquitous concept in the blockchain industry. This is for a good reason.

Historically, the common understanding of what a smart contract is, has been established in the technological context of Ethereum. Ethereum has two types of accounts (we ignore here the latest ERC-4337 addition of so called "smart accounts"):

- externally owned accounts (EOA)
- smart-contract accounts (SCA)

The model of Ethereum was heavily inspired by object-oriented programming ideas, so the above separation can be explained in the following way:

- the blockchain is a computer
- EOAs represent users (and are just cryptographic key-pairs, used for signing transactions)
- SCAs represent programs (and operate as if being standalone mutable objects)

Inside a SCA data and code operating on this data are bundled together, i.e. exactly like you would have in a singleton object in any object-oriented programming language. Because public methods of a SCA form an API that other SCAs (and transactions) can call, it can also be seen as a small "server" operating on-chain.

However in most object-oriented programming languages, classes and instances are separated, so that classes contain code, while objects contain data. This way, a class serves as a template and several instances of it (objects) can be created, all of these objects sharing the same behavior. This abstraction is

not properly reflected in the Ethereum model. This problem can be easily fixed, though - it would be sufficient to detach code from objects, i.e. have objects and classes as separate concepts in the VM. But when you follow this idea, the concept of a "smart contract" does not fit to the situation anymore.

This is exactly what we do in the Khalani blockchain:

- code is in modules
- data is in cells
- every cell is really an object - its "class" is the type it points to

Because every cell points to its type, cells correspond to objects in the object-oriented model. Features of a cell are defined in the module which defines the cell's type.

In summary: smart contracts from Ethereum can be ported to the Khalani blockchain - every smart contract would then correspond to a cell. We just do not use the term "smart-contract" for a cell because the code defining the structure and behavior of the cell is not part of the cell (instead it is in the corresponding module). We also do not want to use the term "smart-contract" meaning the code inside of a module, because this could create even bigger confusion for people used to Ethereum model (because in Khalani the closest thing to Ethereum smart contracts are really (shared) cells).

2.2.5 Cell usage patterns

The blockchain cell-based model is very flexible and it is not desirable (or even possible) to enumerate up-front all the software patterns that developers will come up with over time. But we wanted to mention three of such patterns:

- service
- asset
- mandarin

A **service** is a pattern that effectively resembles smart-contract accounts from Ethereum: there is an on-chain object sitting at a fixed well-known address and offering a public interface to be called by other on-chain actors (objects and accounts). And example of such service could be a DEX, a bridge, a name-registration service, an oracle or a facade to on-chain economical game.

An **asset** pattern is for creating generally "passive" resources which can be traded/purchased: cryptocurrencies, NFTs, wrapped coins, soul-bound tokens. An built-in cryptocurrency of Khalani is also materialized using asset pattern.

A **mandarin** pattern is the one we promote for intent-based processing. We introduce it later in the document.

2.2.6 Consensus

We have not decided yet which consensus protocol will be used in the Khalani blockchain. Most likely this will be a variant of proof-of-stake, which takes into account the native concurrency of the UTXO model.

The UTXO model is a hypergraph, where transactions are hyper-edges. The computational dependencies between transactions have the structure of a partial order. What the consensus protocol has to do is to achieve byzantine agreement across blockchain nodes in regards to this partial order. An example of such protocol is Mysticeti:
`https://sui.io/mysticeti`

2.3 Delegation

Generally speaking, on a blockchain everything is permission-based. In a basic model of accounts (like on Ethereum) access rights are implied by 3 rules:

- only the transaction sender can transfer ether out from his account
- only the smart contract itself can read/write its instance variables
- only the smart contract itself can transfer ether out from its balance

These rules seem simple and logical but they turn out to be quite limiting when it comes to implementing delegation patterns.

Delegation is a family of patterns based on a simple common idea:

> Alice has permission to perform action X. She has however some reasons why she does not really want to perform X herself. Rather, she wants to ask Bob to perform X on her behalf. For this to be possible, the permissions of Alice are delegated to Bob.

Ideally we want the delegation mechanism to be maximally restrictive, so that the scope of Bob using Alice's permissions is maximally narrow (so that Bob cannot abuse the delegation mechanism against Alice's will).

The lack of native mechanism of delegation in Ethereum leads to various troubles, generally pushing developers to creating workarounds like this one:
`https://eips.ethereum.org/EIPS/eip-2612`

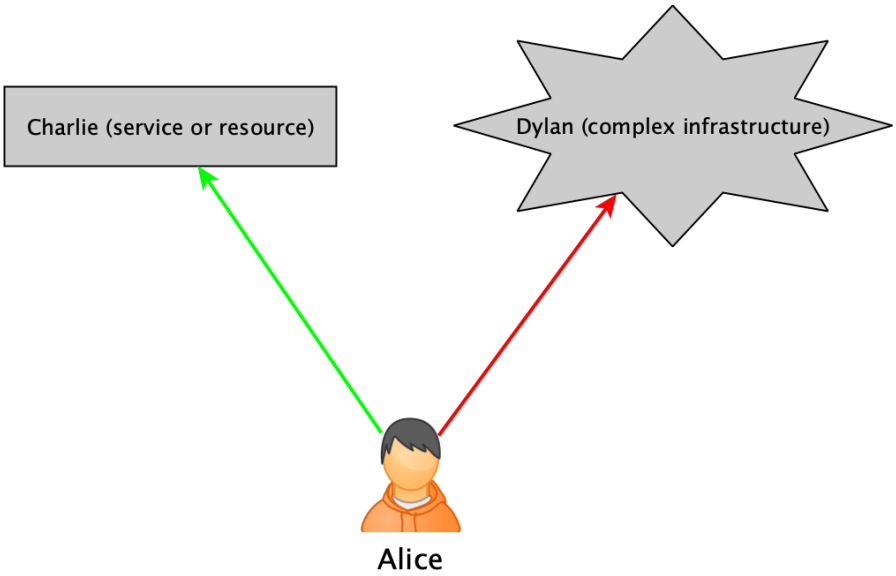
In the meantime, some attempts to introduce general-purpose universal delegation mechanism are underway (ERC-4337 and EIP-7702).

Let's look closer at the example situation we sketched in the Introduction chapter:

> Say we have Alice (user), Bob, Charlie and Dylan (on-chain services). Bob provides some complicated functionality which involves further interaction with Charlie and Dylan. Alice has no rights to interact with Dylan, so she is forced to use Bob as a privileged intermediary. Charlie recognizes Alice as a user with certain access rights. Bob cannot directly call Charlie because Bob has no rights to do so. Hence Bob would like to interact with Charlie on behalf of Alice

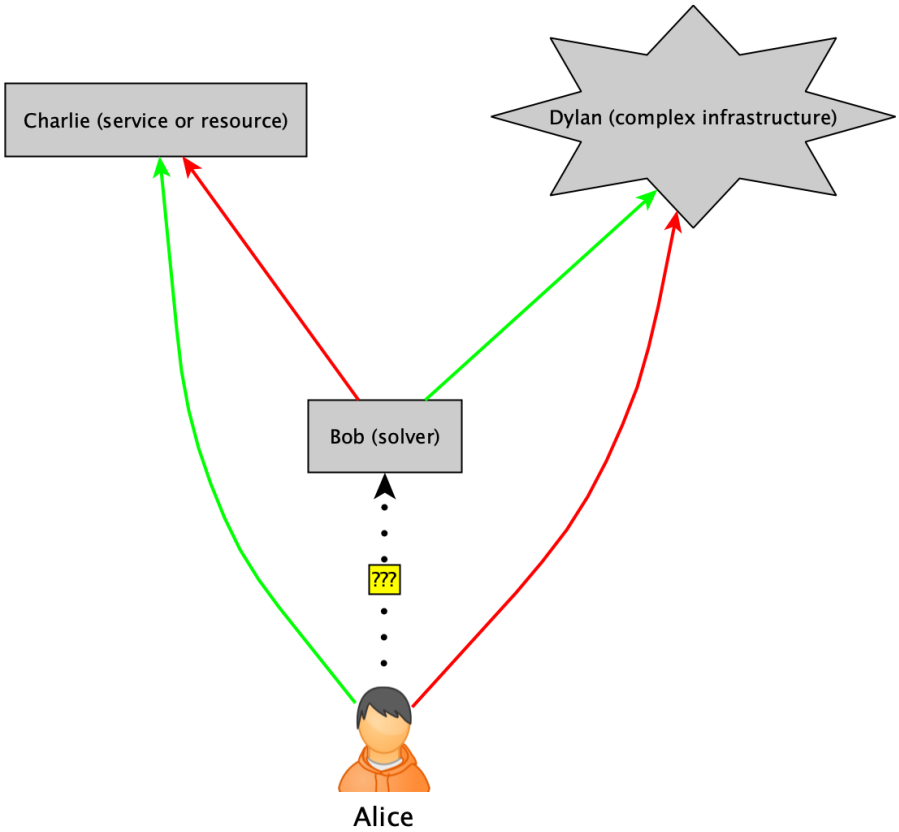
On the pictures below we use green arrows to denote "has access" and red arrows to denote "access denied".

So we start with Alice who has a problem:

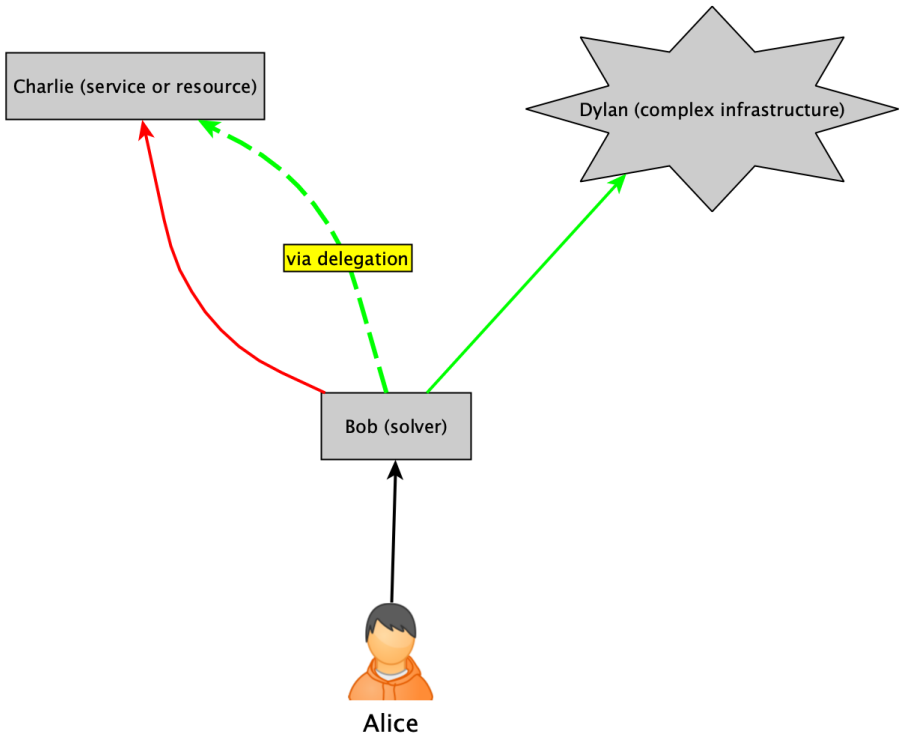


Alice want to achieve certain business goal. However, for this to happen, Alice would need to use both Charlie and Dylan to achieve her goal. Charlie is a service or resource which Alice has access to. Dylan is a complex or access-restricted infrastructure, which is beyond Alice' reach.

Alice wants to solve her problem. She found a helper service Bob (hence we named it "a solver"), who can help her with the complicated task of accessing Dylan infrastructure:



The missing element here is the ability to **delegate** Alice's access rights. She wants to temporarily lend Bob her ability to access Charlie, so that Bob can solve her problem.



The delegation mechanism is potentially very powerful. In particular it is a crucial element of the infrastructure needed for intent-based processing (which we explain later, in the **Intents** chapter). At the same time, any such mechanism is extremely dangerous, because by its very nature it is there to bypass access-rights limitations, hence malicious players can utilize this as a a universal fraud tooling.

In Khalani, we attempt to get best of both worlds: maximal safety and maximal flexibility. Therefore, we do not follow the Ethereum strategy to introduce delegation. Instead, our delegation model is based on a variant of the capability pattern. We allow Alice to form a so called "speculative handle" to represent her rights to access Charlie. Alice can pass such a handle to Bob so to execute the delegation of privileges. However, this delegation is non-exclusive (i.e. can be given to several solvers) and it also can be revoked.

2.4 Pub-sub, timers and gas reserve

2.4.1 Pub-sub on-chain

Publish-subscribe is a well established programming pattern:

`https://en.wikipedia.org/wiki/PublishâĂŞsubscribe_pattern`

In most mainstream blockchains a smart contract can emit events, but the observers can only be off-chain agents. Such version of pub-sub is of course useful, but it has severe limitations. For example if used for the solvers reputation system (for the context what are solvers reputation systems and why they are

needed see below, where we discuss intents) then the off-chain relayer is needed so to pass events to an on-chain reputation registry. This relayer will inevitably happen to become a centralization point, undermining the trust. This would be especially troublesome given the fact that the solvers reputation system is all about establishing the trust levels for solvers.

The escape from such circular problems is simple when a blockchain supports on-chain pub-sub mechanics. This means:

- any smart contract can announce events
- any smart contract can subscribe/unsubscribe to events
- subscription involves handler registration
- when an event is emitted, all the subscribing handlers are invoked

From the developer's perspective this can be captured as follows:

```
type EventHandler<T> = fn(String, data: T);

pub trait EventEmitter<T> {
    fn subscribe(&self, symbol: String, handler: EventHandler<T>);
    fn unsubscribe(&self, symbol: String);
    fn emit(&self, symbol: String, data: T);
}
```

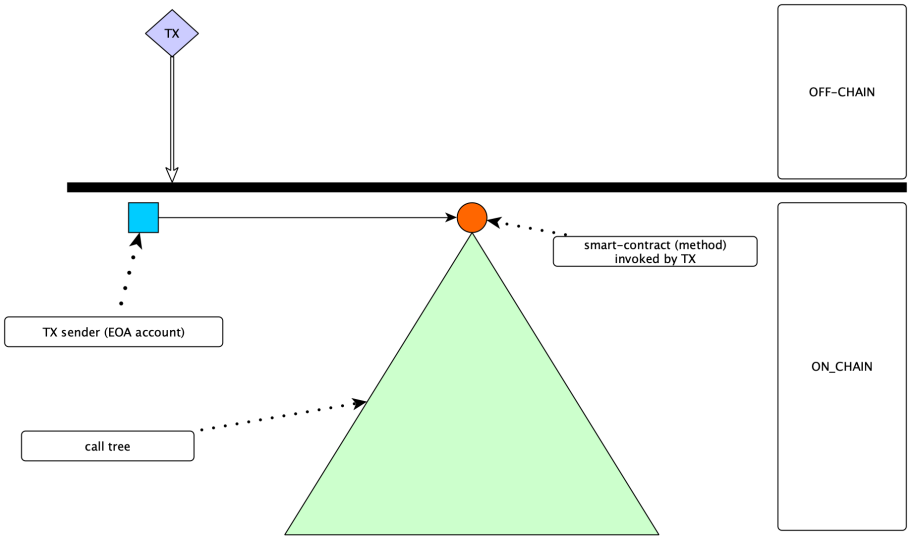
However the semantics of on-chain events is more tricky compared to traditional Ethereum-like event systems with off-chain observers only. The new problem here is "who pays gas for the handler's execution". We discuss this problem in the next chapter.

2.4.2 Gas reserve

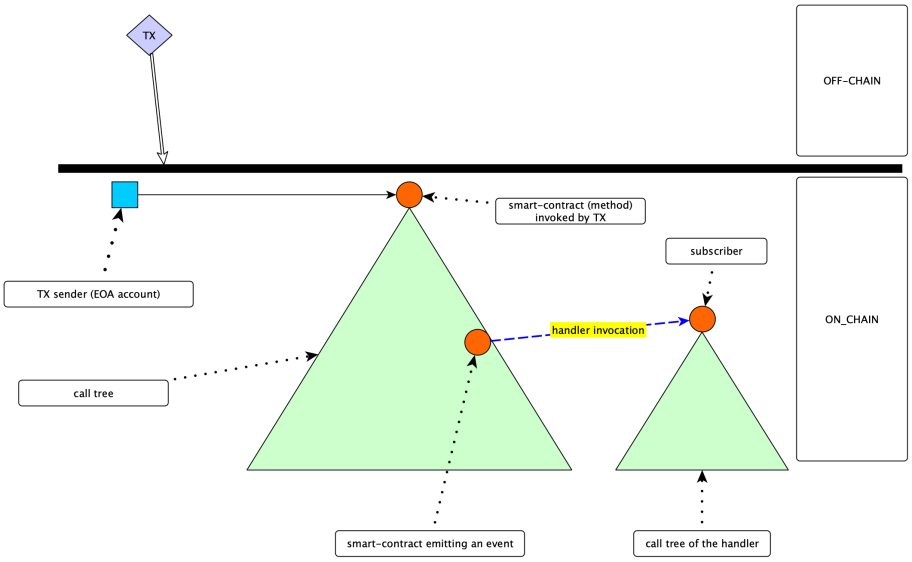
Implementing on-chain pub-sub mechanism on any blockchain inevitably leads to the following problem: when an event is emitted, who is going to pay for the gas cost of invoked handlers?

Obviously we cannot charge the event emitter - this would be against the very idea of events. We basically want the subscriber to handle costs of handlers he registered via subscriptions. Now it becomes clear why for example Ethereum does not support on-chain pub-sub: the desired gas charging semantics would fundamentally break the gas model of Ethereum.

Without pub-sub, the payment model is simple: the whole call tree gas cost is covered by the balance of the sender account:



With pub-sub present, the situation gets more tricky. The invocation of an event handler is not really a "call", because it is the pub-sub machinery doing this call. Hence every handler invocation spawns a separate call tree.



Who should cover the gas cost of the handler call tree is not entirely obvious. It could be the:

- the subscriber
- the sender of the transaction who invoked subscription

Both approaches turn out to be somewhat disappointing. Big part of the problem is that gas prices are fluctuating over time.

What the system should do if the gas cost of the handler execution is not sufficiently covered? The only reasonable option is to rollback the whole handler invocation. Unfortunately, such an approach makes the whole pub-sub abstraction too non-deterministic to be useful.

A safer deterministic design can be obtained if we combine the following rules:

- "best effort" subscriptions and "guaranteed" subscriptions are distinguished
- "guaranteed" subscriptions are one-off (i.e. get cancelled after one handler call)
- "guaranteed" subscriptions are covered with gas reserve (see below)
- the handler registered during a guaranteed subscription must follow the static gas regime (see below)
- explicit **gas reserve** is maintained by every smart contract (or Gas is turned into separate cryptocurrency)
- static gas estimation is supported (for sufficiently limited sub-language of the native smart-contracts language)

2.4.3 On-chain timers

Timers are almost like pub-sub, but it is the time that triggers the invocation of a handler. Of course the concept of time must be somehow available on-chain. In the simplest form the time could be represented as the height of the current block. Covering handlers gas cost for timers leads to similar problems as it was in case of pub-sub.

2.4.4 Transactional approach to pub-sub and timers

The traditional approach to covering gas cost is: the execution cost of the whole call tree is paid by the transaction sender.

However - as we argued above - this model breaks down when calling handlers: Someone else is to pay for the handler. It leads to an interesting question: if the call tree of transaction A invoked handler B (effectively another call tree) and during the execution of call tree B the corresponding gas/cryptocurrency reserve has been exhausted - what should happen? Of course the execution of call tree B has to stop immediately. But the state of the blockchain cannot stay inconsistent - so we need to do a rollback. But because an event emitter must be shielded from any errors during the execution of handlers, it leads to the conclusion that we need to rollback call tree B only, while the call tree will stay intact.

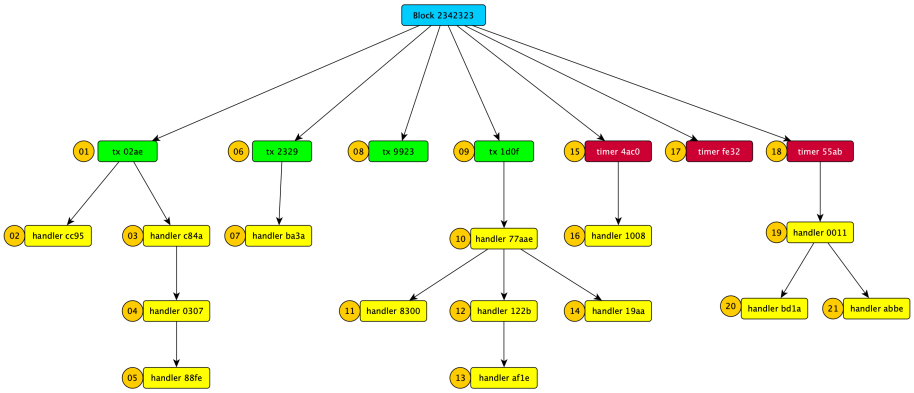
The simplest way to achieve this isolation and to retain the original gas semantics is to assume that *handler invocation is executed as a separate transaction.*

In other words, in such a blockchain we would have 4 types of transactions:

- (external) on-chain code deployment

- (external) on-chain function call
- (internal) handler invocation
- (internal) timer invocation

External transactions are sent to nodes by clients. Internal transactions are added by the VM on-the-fly during the execution. The only remaining question is the ordering. The outcome of the consensus protocol is the linear ordering of blocks, and linear ordering of (external) transactions inside of blocks. When the block is executed, necessary internal transactions will be created and inserted into the original sequence of external transactions. This process can be illustrated is the following tree:



following tree:

1. Colors:
   (a) green - external transactions (ordered by the consensus protocol)
   (b) yellow - pub-sub handlers invocations
   (c) red - timers handlers invocations
2. While executing any transaction, the VM accumulates handler invocations launched by event broadcasters. Every such handler invocation will be then executed in a separate (internal) transaction.
3. Because any on-chain code can emit events, this means that any transaction (internal or external) can spawn new transactions by emitting events. This leads to a tree where nodes are transactions and edges are handler invocations. Because the whole block has the same blockchain time, timer handlers are placed at the end of a block (after all the external transactions). Their ordering should be implied by the ordering of respective subscriptions.
4. Ordering of child nodes is implied by the order of handler invocations:
   (a) for different ‘emit()‘ invocations, the order of resulting internal transactions follows from the ordering of emit() calls (on-chain code is executed sequentially)

(b) for a single ‘emit()‘ invocation, the order of resulting internal transactions follows from the order of subscriptions

5. Ordering of all transactions in a block is achieved as the variant of depth-first-search traversing of the tree, where every parent mode is visited before its child nodes.

6. This algorithm ensures the following properties:

   (a) The ordering of external transactions as achieved by the consensus protocol is never changed.

   (b) If (internal or external) transaction T1 precedes transaction T2 then all event handlers spawned by T1 will be executed before T2.

   (c) Gas for every transaction (internal or external) is charged separately.

   (d) Execution of event handlers (including any errors) is not influencing the execution of the transaction that emitted events which spawned these handlers. In particular, rollback of a handler transaction will not influence the transaction where the corresponding event originated.

2.4.5 Khalani approach

The exact way timers, pub-sub and gas reserve will be accessed at Axi level is not decided yet. Nevertheless, we want all 3 mechanisms to be available.

Most likely we will not introduce Gas as a separate cryptocurrency, because this may lead to creation of a massive gas trading market, which in turn could undermine the principle reason for gas existence (it is there to achieve supply-demand balance around the processing power of the blockchain). We are leaning to a more limited gas reserve semantics, where an object can accumulate private, non-transferrable gas reserve for own use only and the gas price to be paid when buying gas for the reserve is $p\%$ bigger than the gas price of the current block. The parameter p fluctuation would then be subject to decentralized management by validators voting.

Also, because we want provability and this involves proving behaviors that depend on events, the static-gas sublanguage of Axi will be delineated. The VM will be able to compute the gas cost of a function written in the static-gas sublanguage of Axi by static analysis. Static-gas sublanguage can be then used to implement a handler that is part of critical functionality, so that a correct gas reserve can be allocated upfront.

2.5 Intents

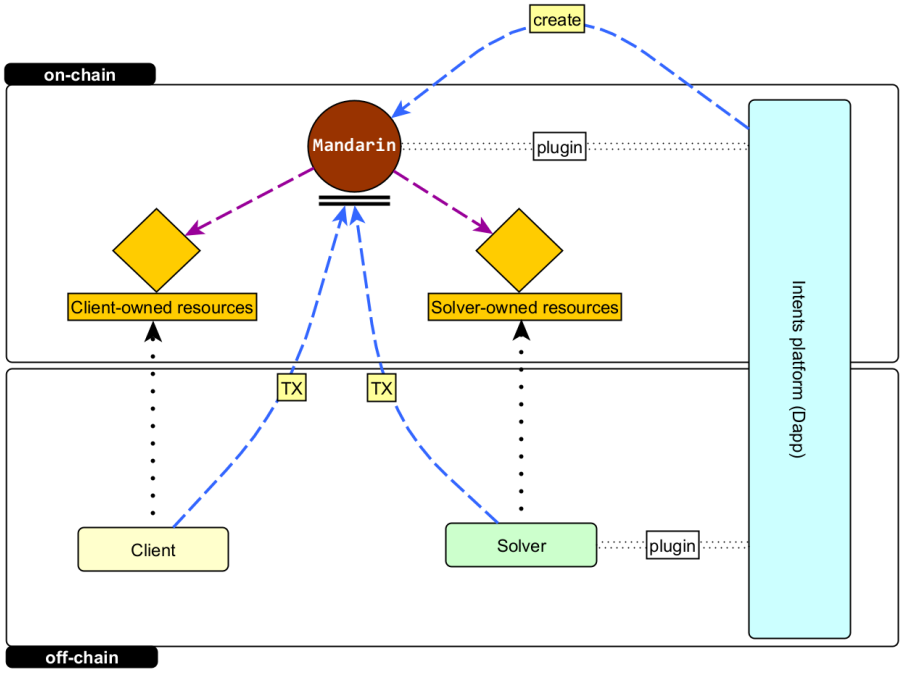
2.5.1 Intent-driven processing

TODO

TODO

Table 2: Concepts related to intent markets

DEX concept	Corresponding concept in i-markets
order	intent
orderbook	intentbook
order execution	filling
order type (limit order, market order, stop order...)	intent type



TODO

Appendices

A Overview of Axi

Axi (pronounced "Axee") is the smart contract language of Khalani. It is a purely functional programming language with a strong, static, linear type system and support for formal reasoning about programs in classical higher-order logic. Among mainstream programming languages, it is most similar to Haskell (in spirit) and OCaml (syntactically). It was also heavily inspired by proof

assistants based on type theory, like Coq, Lean and Agda.

In this appendix we will have a brief, not too technical look at the philosophy and language features of Axi, together with a few simple code examples.

A.1 Philosophy

Axi's architecture is unique in that the language is split into two strictly separate layers: the programming layer and the logic layer.

The Programming Layer of Axi is a purely functional language based on System F with quantitative typing (linear, affine, relevant and unrestricted types) and algebraic data types. It aims to make programming pleasant and comprehending smart contract code easy, while providing essential guarantees to all on-chain code:

- **Type safety**. Well-typed programs can't go wrong, i.e. they can't crash (unless they run out of gas), can't perform nonsensical operations like adding a number and a string, and can't result in any other runtime type error.
- **Memory safety**. Well-typed programs can't result in memory errors, like writing to a random memory location, nor can they have memory leaks.
- **Resource safety**. Well-typed programs can't access resources they don't have permission to, can't delete tokens that are not supposed to be deletable and can't create money out of thin air.

The Logic Layer of Axi is a language for reasoning about Axi programs in classical higher-order logic. It allows expressing arbitrarily complex mathematical properties, thus giving programmers the ability to precisely specify the business logic and behaviour of their programs. These specifications can then be either tested with random inputs before deployment, checked for concrete inputs at runtime, or proved for all possible inputs using logic. The last of these, proving, provides guarantees much more powerful than those of the type system alone, while being as credible and trustworthy as the type system. The use of the logic layer is entirely optional, i.e. programmers are never forced to engage with any part of it, although competition will encourage programmers to do so in the hopes of outcompeting other programmers' smart contracts by providing stronger guarantees.

Strict separation. In Axi, logic can reason about programs, but programs cannot manipulate logical entities, like theorems or proofs, at runtime. This one-way interaction keeps the language relatively simple, while still making it possible to express almost all functional programming idioms and all possible mathematical properties and their proofs.

Guarantees about guarantees. We not only want Axi to provide strong guarantees about its programs, but we also want to provide strong guarantees about Axi itself. We plan to use the Coq proof assistant to formally verify significant portions of Axi's metatheory, including the aforementioned properties of type and resource safety, as well as the consistency of Axi's logic. We are also considering formal verification of the reference implementation of Axi. The goal of these efforts is to ensure that the guarantees that Axi provides are trustworthy and will stay so forever, as long as the consensus protocol is followed.

A.2 Language features

In this section we will take a closer look at Axi's language features.

A.2.1 The Programming Layer

Axi is, at its core, a purely functional programming language. It thus supports all the usual features of such languages:

- **Purity amd immutability**. In Axi, all values are pure and immutable – they cannot be null, cannot throw exceptions and once created, they cannot be changed. This facilitates reasoning about and comprehending code, thus rendering Axi smart contracts transparent and auditable.
- **First-class, higher-order and anonymous functions**. Functions are first class citizens. They can be passed as arguments to other functions, returned from other functions as results and created on the fly. Axi encourages a programming style in which data is processed by pipelines built from basic building blocks like mapping, reducing, filtering and so on.
- **Algebraic data types and exhaustive pattern-matching**. Instead of clunky boolean flags and cascades of ifs, Axi makes it easy to model the problem domain with disjoint unions and define functions on them by pattern matching and explicitly handling all possible cases.
- **Strong static type system**. Axi's type system makes it easy to work with high-level abstractions, like merkle trees or state machines, instead of low-level primitives, like integers or byte arrays. It is lightweight, not placing too many demands on the programmer, but still catches the most egregious errors – it can easily protect programmers from confusing with each other such objects as an integer, a string, an address, a transaction hash, a token amount and a cryptographic secret.
- **Polymorphism and traits**. Axi's type system features polymorphism (also known as "generics") as well as traits inspired by Haskell's type classes (but named after Rust's traits). These features make it easy to implement data structures and functions that work for all types that support a certain behaviour, for example a hash map that works for all types of ERC-20-compliant tokens bridged from an external blockchain.

Quantitative typing. The standard features of purely functional programming is not everything that Axi has to offer. The killer feature, as far as the programming layer is concerned, are quantitative types, which allow the type system to enforce correct resource usage discipline. This aspect of Axi's type system was inspired by linear logic and its cousin, quantitative type theory, and is an improvement upon an analogous feature of the type system of Move, the programming language of the Sui and Aptos blockchains.

In an Axi program, every variable is assigned its *quantity*, which can be either: zero, one, at most one, at least one, unrestricted. The variable's quantity determines how many times it can be used in the program, with too many or too few uses resulting in a type error. For example, the typical variable holding an NFT will have a quantity of one, forcing the program to do something with the

NFT before it terminates: maybe send the NFT to the transaction's originator, maybe trade it for something else and handle that something else properly according to its quantity, or maybe send it to somebody as a gift. What the program cannot do in such a situation is forget the NFT and do nothing with it, effectively making it "lost" in the guts of the chain. It also cannot copy the NFT and have two instead of just the original one.

To make the programmers' lives easier, every type is also assigned its "default" quantity. Every variable will be assigned the default quantity of its type unless the programmer decides otherwise. An important piece of terminology: a type whose default quantity is unrestricted is called a *data type*, whereas a type whose default quantity is either one, at least one or at most one, is called a *resource type*. There are no types whose default quantity is zero. Elements of data types are called data, whereas elements of resource types are called resources.

Traits. A trait is a way of representing an interface satisfied by a type. The definition of a trait consists of a bunch of declarations of constants and functions. The programmer can make a type an *instance* of a trait by implementing the trait's constants and functions for this type.

A.2.2 The Logic Layer

Axi's logic layer consists of typed classical higher-order logic with equality whose domains of discourse are the types from the programming layer. More precisely, this means the logic has:

- **Propositional connectives and constants**. Standard propositional connectives ($\Longrightarrow$, $\wedge$, $\vee$, $\Longleftrightarrow$, $\sim$) and constants ($\top$, $\bot$) can be used to build up complex propositions from simpler ones.

- **Quantification over terms and types**. Axi supports universal and existential quantification. We can quantify over terms of a given type ($\forall x : A, \dots$) as well as over types themselves ($\forall A : \texttt{Type}, \dots$), no matter whether we're dealing with data, resources or programs.

- **Higher-order quantification**. We can also quantify over logical entities like propositions ($\forall P : \texttt{Prop}, \dots$), predicates ($\forall P : A \to \texttt{Prop}, \dots$), relations ($\forall R : A \to A \to \texttt{Prop}, \dots$), logical connectives ($\forall C : \texttt{Prop} \to \texttt{Prop} \to \texttt{Prop}, \dots$) or even quantifiers themselves ($\forall Q : (A \to \texttt{Prop}) \to \texttt{Prop}, \dots$). This ability of logic to talk not only about the underlying programming language, but also about itself, gives it great expressive power and allows it to make use of all of mathematics when reasoning about the behaviour of programs.

- **Equality**. The built-in equality relation ($=$) makes it possible to precisely reason about programs' input-output behaviour.

Proving is carried out in a proof language based on the Curry-Howard correspondence: proofs of implications and universal quantifiers are like functions, proofs of existential quantifiers and conjunctions are like pairs, proofs of disjunctions are like disjoint union constructors, etc. Negation and biconditional are handled as in intuitionistic logic - instead of being primitives, negation ($\sim P$) is a shorthand for an implication into falsehood ($P \implies \bot$), whereas

biconditional ($P \iff Q$) is a shorthand for a conjunction of two implications ($(P \implies Q) \wedge (Q \implies P)$). The principle of *proof by contradiction*, which allows to prove $((\sim P \implies \bot) \implies P)$ for any proposition P, makes the logic classical and thus intimately familiar to all mathematicians.

Equality is handled as in Martin-Löf Type Theory – proved by reflexivity and eliminated with the so-called J rule, which is a principle that allows to substitute equals for equals in the remaining part of the proof. Because proofs are hard to write using these primitives, Axi provides a convenient feature called *chaining*, which allows reasoning with long chains of equations by writing down and justifying each step separately, similarly to how equality is usually handled with pen and paper. Besides equality, this feature also supports implications, biconditionals and other transitive relations.

One novel feature that distinguishes our proof language is that the proofs can be presented in two styles: either in a *proof term style*, similar to how proofs look in Agda and Idris (as well as Coq's and Lean's manually written proof terms), or in a *tactic style*, similar to how proofs usually look in Coq or Lean. The difference between Axi and languages that also support both styles, like Coq and Lean, is that in these languages each style is its own separate proof language, whereas in Axi there is a single proof language which, thanks to a clever parser, can be formatted either like terms or like tactics. This feature will make it easier to accomodate computer scientists and mathematicians familiar with either of the aforementioned languages.

Concepts – traits in logic. Last but not least, the logic can not only talk and reason about the traits we met in the programming layer – it also introduces the richer and more powerful construct called *concepts*. Concepts extend traits with specifications, i.e. properties expected to be true of the trait's constants and functions. When implementing a concept, the programmer must also provide proofs of the specifications. Thus concepts let programmers precisely define what is expected of trait implementations.

B Code examples

In this section we will see some code examples that demonstrate how Axi works in practice. Note that Axi is not implemented yet, so that the code examples are only an approximation of what we intend to achieve, instead of being a demonstration of an existing language.

Example 1. *An example program – list functions*

```
data List (A : Type) : Type where
  nil
  cons (h : A) (t : List A)

app @(A : Type) (l1 l2 : List A) : List A =
  match l1 with
  | nil      => l2
  | cons h t => cons h (app t l2)

map @A @B (f : A -> B) : List A -> List B
| nil      => nil
| cons h t => cons (f h) (map f t)
```

The above example demonstrates the basics of Axi programming. In lines 1-3 we define an algebraic data type called `List`. The type is parameterized by a type `A` and has two constructors: `nil` is the empty list and `cons h t` is the list with head `h` and tail `t`. Both annotations `: Type` are necessary, because we need to indicate that both `A` and `List A` are unrestricted types, as opposed to linear, affine or relevant (we will see these in later examples).

In lines 5-8 we define a function called `app` which concatenates two lists. It takes three arguments: a type `A` and two lists `l1` and `l2` of type `List A`, and returns a result of type `List A`. The type argument means that the function is polymorphic. It is also marked with the symbol `@`, which means it's an *implicit* argument, i.e. when calling the function it can be omitted and will get automatically inferred. The body of the function is simple: we pattern match on the argument `l1`. If it's `nil`, we return `l2`. If it's of the form `cons h t`, the result's head is `h` and its tail is computed by the recursive call `app t l2`.

In lines 10-12 we define another function, called `map`, which applies a function `f` to every element of a list. This time, however, we used an abbreviated syntax. The two type arguments `@A` and `@B` are marked as implicit, but we don't write `: Type`, which gets inferred automatically. The argument `f` and its type are written out in full. The last argument, of type `List A`, and the result type `List B`, appear after the last `:` as the type `List A -> List B`. Because of this, it is understood that in the next, pattern matching on the input of type `List A` follows.

Example 2. *Quantitative typing*

```
// ExampleCoin : Type1 is a resource type that we'll define later.

coinForgery (x : ExampleCoin) : ExampleCoin = coinMerge x x
//           ^                                            ^ ^
//           |                                            | |
//           \-------------------------------------\      | |
//                                                 |      | |
// Type error: duplicate use of the resource 'x' -/       | |
// of type 'ExampleCoin : Type1'                           | |
// First use ------------------------------------------/ |
// Second use -----------------------------------------------/
// Hint: remove the duplicate use
// or declare the type to be in 'Type+' or 'Type' to allow it

coinBlackhole (x : ExampleCoin) : Unit = ()
//             ^                         ^
//             |                         |
//             \--------------------\    |
//                                  |    |
// Type error: unused resource 'x' -/    |
// of type 'ExampleCoin : Type1'         |
// Hint: use the resource in body -------/
// or declare the type to be in 'Type?' or 'Type' to allow it
```

The above example demonstrates the kind of resource discipline enforced by the type system. We will define the type `ExampleCoin` and the function `coinMerge` in the next example. For now it only matters that `ExampleCoin` is a resource type, i.e. `ExampleCoin : Type1`.

First we try to define a function called `coinForgery`. It takes an argument of type `ExampleCoin` called `x` and returns an `ExampleCoin` by calling the function

`coinMerge` with boths its arguments being `x`. Now, `ExampleCoin` is a linear type, i.e. by default variables of this type must be used exactly once. Because `x` is of type `ExampleCoin` and nothing indicates otherwise, this means that the function `coinForgery` must use `x` exactly once. However, it uses it twice when calling `coinMerge`, which is a type error.

This error is caught by the type checker, which complains that `x` was used too many times. The error message also hints at some possible solutions: use `x` less (the only solution for ordinary programmers) or redefine the type `ExampleCoin` to allow duplicate uses (this solution is only available to the type's implementer). The fact that the type system was able to catch this error is crucial, because if the function `coinForgery` passed the type checker, it would allow programmers to create unbounded amounts of `ExampleCoin`, causing hyperinflation, which most likely was not the intention of this type's creator.

The second function, called `coinBlackhole`, takes an argument `x` of type `ExampleCoin` and returns a value of type `Unit`, which is a type with only one element `()`. The `Unit` type is returned to indicate that the function's result "doesn't matter". Because the type of `x` is linear, `coinBlackhole` must use `x` exactly once. However, it doesn't use it at all, which is a type error.

This error is caught by the type checker, which complains that `x` was not used. Similarly to the previous error message, two solutions are offered: use `x` or make the type `ExampleCoin` affine so that it allows zero or one usages. The fact that this error was caught by the type system is very important, because `ExampleCoin`'s creator did not intend for values of this type to be deletable.

Example 3. *Coin trait*

```
trait Coin (A : Type1) where
  coinAmount : A -> Nat * A
  coinZero   : A
  coinMerge  : A -> A -> A
  coinSplit  : Nat -> A -> Option A * A

record ExampleCoin : Type1 where
  amount : Nat

instance Coin ExampleCoin where
  coinZero : ExampleCoin =
    record where amount = 0

  coinMerge (c1 c2 : ExampleCoin) : ExampleCoin =
    record where amount = c1.amount + c2.amount

  coinAmount (c : ExampleCoin) : Nat * ExampleCoin =
    let n : Nat = c.amount in
      (n, record where amount = n)

  coinSplit (askedAmount : Nat) (c : ExampleCoin) :
    Option ExampleCoin * ExampleCoin =
      let n : Nat = c.amount in
        if askedAmount <= n
        then
          ( some (record where amount = askedAmount),
          , record where amount = n - askedAmount)
          )
        else
          (none, record where amount = n)
```

The above example shows the definition of the `Coin` trait, which could be one way of capturing the commonalities of user-defined coins. Then we implement the type type `ExampleCoin` from the previous example and demonstrate how to make it an instance of the `Coin` trait.

In lines 1-5 we define the trait `Coin`. It takes a parameter `A : Type1`, i.e. this trait can only be implemented by resource types. The trait has one constant and three functions. `coinZero` represents a coin with denomination zero. `coinMerge` merges two coins. `coinSplit` splits one coin into two according to the amount given in its first argument, with the possibility of failure (if the amount asked is too big) indicated by `Option` in the return type. `coinAmount` is perhaps the most mysterious – it returns the denomination of the coin, but also a new coin. This is a pattern that we will see many times to come. If `coinAmount` returned just the amount, that would destroy the coin, because it must be used exactly once (because its type is linear), and checking the amount counts as a single use! Therefore, `coinAmount` must not only return the amount, but also a ‘`new” coin, which we expect to be the same as the` ‘old” coin (but this is not guaranteed; later we will see how to guarantee this fact using logic).

In lines 7-8 we implement the type `ExampleCoin`. It is a record type with a single field `amount` which is a natural number (i.e. a non-negative number which can be arbitrarily large). `ExampleCoin` is a resource type, as marked with the annotation `: Type1`.

In lines 10-27 we implement the trait `Coin` for the type `ExampleCoin`. Note that we don’t need to implement the trait’s constants and functions in the same order they appear in the trait definition. In lines 11-12 we implement `coinZero` – it’s a record (of type `ExampleCoin`) whose `amount` field is `0`.

In lines 14-15 we implement `coinMerge`. It takes two arguments of type `ExampleCoin` and returns an `ExampleCoin` which is a record whose `amount` field is the sum of the `amount`s of its arguments.

In lines 17-19 we implement `coinAmount`. It takes an argument `c` of type `ExampleCoin` and returns a tuple whose first component is a natural number and whose secnd component is an `ExampleCoin`. First, in line 18, we consume the coin `c` by peeking at its `amount` field, which then gets named `n`. In line 19, we build the return tuple – the first component is just `n`, whereas the second component is a record whose `amount` field is `n`. Note that the latter is exactly the same thing as the initial coin `c`, but we cannot directly return `c` as the secnd component, because it must be used exactly once and we have already used it when checking its `amount`.

Last but not least, in lines 21-30 we implement `coinSplit`. It takes two argument, `askedAmount` and `c` of type `ExampleCoin`. First, in line 23, we consume `c` to check its amount and store it in `n`. Then on line 24 we check if `n` is big enough for the split to succeed. If it does, on lines 26-28 we return a tuple whose first component is a coin whose amount is the `askedAmount` (wrapped in `some` to indicate success) and whose second component is a coin whose amount is `n` minus the `askedAmount` that went to the other coin. If `n` is not big enough, we return a tuple whose first component is `none` (to indicate failure) and whose second component is a coin whose amount is `n`, i.e. a coin that is the same as the original coin `c`.

Example 4. *A simple proof*

```
1  theorem app-assoc :
2    forall (A : Type) (l1 l2 l3 : List A),
3      app (app l1 l2) l3 === app l1 (app l2 l3)
4  proof
5    pick-any A l1
6    induction l1 with
7    | nil =>
8      pick-any l2 l3
9      refl
10   | cons h (t & ind IH) =>
11     pick-any l2 l3
12     chaining
13       === app (app (cons h t) l2) l3
14       === cons h (app (app t l2) l3)
15       === cons h (app t (app l2 l3))  by rewrite IH
16       === app (cons h t) (app l2 l3)
17 qed
```

The above example shows a simple theorem and its proof. The theorem's name, `app-assoc` (which is a kind of mnemonic for "(list) append is associative") is in line 1. Its statement, in lines 2-3, reads as follows: for all types `A` and for all `l1`, `l2`, `l3` which are of type `List A`, the list `app (app l1 l2) l3` is equal to the list `app l1 (app l2 l3)`.

The proof is on lines 4-17. At any point in the proof, Axi keeps track of all the *goals* that need to be solved to finish the current proof, and also of the *context* of these goals, which is the list of all local objects and assumptions we can use in the proof. When we start the proof with the keyword `proof`, we start with a single goal that is the same as the theorem statement. The code on line 5 could be read "let `A` be any type and let `l1` be any list of `A`s". At this point the goal becomes `forall (l2 l3 : List A), app (app l1 l2) l3 === app l1 (app l2 l3)`. On line 6 we indicate that the rest of the proof proceeds by induction on the list `l1`. The induction has two cases, `nil` and `cons`, corresponding to the two list constructors. This means that we now have two (sub)goals, one for each case.

On line 7 we indicate the start of the (sub)proof for the first goal, i.e. the `nil` case. At this point our current goal is `forall (l2 l3 : List A), app (app nil l2) l3 === app nil (app l2 l3)`, because `l1` was replaced with `nil`, and after line 8 it is `app (app nil l2) l3 === app nil (app l2 l3)`. We finish the first subgoal on line 9. `refl`, which is short for "reflexivity", indicates that this equation holds trivially, just by computing the result on both sides.

On line 10 we indicate the start of the (sub)proof for the second goal, i.e. the `cons` case. On the same line we bind the arguments of `cons`, which we name `h` and `t`, as well as get hold of our induction hypothesis for the tail `t`, which we name `IH`. After line 11, the current goal is `app (app (cons h t) l2) l3 === app (cons h t) (app l2 l3)`, as `l1` was replaced with `cons h t`.

In lines 12-16, we use chaining to finish the goal off. It works like this: we start with the left hand side of the current goal (which must be an equation), which in our case is `app (app (cons h t) l2) l3` (line 13). Then in subsequent lines we write out what this expression is equal to (after the equality symbol `===`) and provide justifications for why this is the case (after the keyword `by`). In our case, on line 14 indicate that our goal's left-hand side is equal to `cons h (app (app t l2) l3)`, but we omit the justification for this. We could have written `by refl`, but this is the most trivial justification possible,

so Axi guesses this is what we mean when we omit a justification. Then, on line 15, we indicate that it is equal to `cons h (app t (app l2 l3))`, justifying it with `by rewrite IH`, i.e. we are making use of our induction hypothesis. This justification is crucial and cannot be omitted, because Axi cannot infer it on its own. Finally, on line 16, we indicate that our expression is equal to the right-hand side of our goal, again omitting the justification, because it is trivial. All in all, the entire chaining proves that `app (app (cons h t) l2) l3` is equal to `app (cons h t) (app l2 l3)`, which is exactly what our goal was. Thus the proof is finished, which we indicate on line 17 with the keyword `qed`.

Example 5. *Lawful coin concept*

```
concept LawfulCoin (A : Type1) extends Coin A where
  coinAmount' (x : A) : Nat =
    let (n, _) = coinAmount c in n

  law coinAmount-spec :
    forall @(x : A),
      let (_, y) = coinAmount x in x === y

  law coinZero-spec :
    coinAmount' coinZero === 0

  law coinMerge-spec :
    forall @(x y : A),
      coinAmount' (coinMerge x y)
        ===
      coinAmount' x + coinAmount' y

  law coinSplit-none :
    forall @(askedAmount : Nat) @(coin coin' : A),
      coinSplit askedAmount coin === (none, coin')
       <->
      (askedAmount <= coinAmount' coin) === false
        /\
      coin === coin'

  law coinSplit-some :
    forall @(askedAmount : Nat) @(coin exact change : A),
      coinSplit askedAmount coin === (some exact, change)
       <->
      (askedAmount <= coinAmount' coin) === true
        /\
      coinAmount' exact === askedAmount
        /\
      coinAmount' change === coinAmount' coin - askedAmount
```

The above example shows a concept called `LawfulCoin`, which extends the trait `Coin` that we met previously. Note this distinction: the trait and its implementation are a part of the programming layer, whereas the concept and its implementation are part of the logic layer.

In lines 2-3, we define a function `coinAmount'` that is part of the `LawfulCoin` concept. This is a feature we have not mentioned before: both traits and concepts, besides declarations, can also contain (auxiliary) definitions. In our case, we need the function `coinAmount'` to avoid using the much more cumbersome `coinAmount` when stating the properties that lawful coins should satisfy. `coinAmount'` takes an argument of type `A` (of which we think as a coin, because

it has the `Coin` trait) and returns only the first output of `coinAmount`, while ignoring its second output.

It might seem that `coinAmount'` ignoring the coin returned by `coinAmount` is a type error (zero uses instead of exactly one), but function definitions that belong to the logic layer play by different rules from ordinary functions. Most importantly, they don't need to adhere to the resource usage discipline. In other words, auxiliary functions defined solely to be used to state theorems can freely delete or copy resources, even it were a type error for an ordinary function to do so. This is not a problem from the point of view of ordinary programs – these "logic-only programs" cannot be run (they can only be reasoned about when proving theorems), so they can never do anything bad to on-chain assets.

Next, in lines 5-34, comes the main part of the concept's definition. In lines 5-7 we declare a law claled `coinAmount-spec` which specifies the behaviour of `coinAmount` – the second component of its output must be equal to its input. This is the guarantee missing from the `Coin` trait that we were expecting to hold. Every implementation of `LawfulCoin` will have to provide a proof that turns this law into a theorem.

Having given a specification of `coinAmount`, all the remaining laws will use `coinAmount'` only. In lines 9-10 we declare the law `coinZero-spec` which says that the amount of `coinZero` is zero. In lines 12-16, the law `coinMerge-spec` says that the amount in a merged coin is equal to the sum of amounts of the individual coins.

In lines 18-24 the law `coinSplit-none` says that `coinSplit` returns `(none, coin')` if and only if the asked amount was not smaller than `coin`'s amount and the coins are equal. However, this only specifies one of the two cases that may happen when calling `coinSplit`, so we need another law. In lines 26-34 the law `coinSplit-some` says that `coinSplit` returns `(some exact, change)` if and only if the asked amount was smaller than `coin`'s amount, the amount of `exact` is equal to `askedAmount` and the amount of `change` is equal to `coin`'s amount minus the asked amount. Together, these two laws fully specify the working of `coinSplit`: if the coin's amount is enough, the coin gets split into a coin with the amount we asked and a coin which contains the "change". If we asked for too much, we just get the original coin back.

Example 6. *Implementation of `LawfulCoin` for `ExampleCoin`*

```
1  instance LawfulCoin ExampleCoin where
2    theorem coinAmount-spec :
3      forall @(x : ExampleCoin),
4        let (_, y) = coinAmount x in x === y
5    proof
6      pick-any x
7      refl
8    qed
9
10   theorem coinZero-spec :
11     coinAmount coinZero === (0, coinZero)
12   proof
13     refl
14   qed
15
16   theorem coinMerge-spec :
17     forall @(x y : ExampleCoin),
18       coinAmount' (coinMerge x y)
19         ===
```

```
    coinAmount' x + coinAmount' y
proof
  pick-any x y
  refl
qed

// The proofs of coinSplit-none and coinSplit-some were elided.
```

The above example shows how to implement `LawfulCoin` for the type `ExampleCoin`. We don't need to reimplement `coinAmount'` or anything like that – this function is defined by the concept and we can use it freely. What we need to do is to replace every `law` from the concept's definition with a `theorem`. It so happens that we implemented `coinAmount`, `coinZero` and `coinMerge` correctly and also our implementations are so simple that the proofs go through just by computation (i.e. `refl`). The only theorems requiring a non-trivial amount of proving would be `coinSplit-none` and `coinSplit-some`, but we omit these proofs because they are more about natural numbers than the concept of `LawfulCoin`.

C Interacting with the ledger in Axi

In this appendix, we will see how Axi programs can interact with Khalani's ledger. First we will discuss a simplified ledger model, which we will use in the remaining code examples, and look at the primitive (buil-in) types and functions that represent it in Axi. Then we will give an overview of the big idea behind Axi's interaction with the ledger, which we might call the "resource lifecycle". Finally we will discuss a slightly longer example of an on-chain service, together with example transactions that set this service up and use it.

C.1 A simplified ledger model and Axi primitives

We will use the word *ledger* to refer to the state of the blockchain. Things on the ledger have their addresses. The three kinds of addresses are *accounts*, *package addresses* and *cell addresses* (or just addresses for short).

Accounts are addresses that act as the identities of external actors. They allow them to interact with the blockchain, similarly to EOAs (Externally Owned Accounts) on Ethereum.

All Axi code deployed onto the ledger is stored in a big codebase, divided into *packages*. Package addresses are means of locating these packages in this codebase. Packages are further subdivided into smaller units, called *modules*, which hold definitions of types, interfaces, constants and functions.

Data and resources on the ledger are stored in *cells*, also called *objects*. Each cell has an address and a type, and stores exactly one value of that type. Cells also have some metadata which won't concern us for now.

Objects are either *owned* or *shared*. Owned objects have exactly one owner who is an account. Shared objects do not have an owner. Shared objects are further subdivided into *mutable shared objects*, which contain resources and can be interacted with by anybody (any account or object) according to the interface set by the package defining the object's type, and *immutable shared objects*, which contain data but not resources and can be read by anybody, but not modified.

The state of the ledger is modified by transactions. Every transaction has a *transaction originator* and a *transaction body*, which is an Axi function that describes how to modify the ledger. Transactions also have some additional metadata which won't concern us here.

Example 7. *Axi primitives for interacting with the ledger*

```
primitive
  // provides access to ledger and transaction metadata
  Ledger : Type1

  // a unique address of a cell or an account
  Address : Type

  // an address that signed the currently executed transaction
  whoami : Ledger -> Address * Ledger

  // create a new cell owned by a specified address
  new : forall (A : Type1), Address -> Ledger -> Address * (A -> Ledger)

  // destroy an existing cell owned by transaction signer
  delete : forall @(A : Type1), Address -> Ledger -> Option A * Ledger

  // reference to a shared cell
  Ref : Type1 -> Type

  // create a new mutable shared cell, accessible by everyone
  share : forall @(A : Type1), Ledger -> Ref A * (A -> Ledger)

  // load contents of a shared cell and then store a new value
  update : forall (A : Type1), Ref A -> Ledger -> A * (A -> Ledger)

  // upcast a reference to an address
  addressof : forall (A : Type1), Ref A -> Address

  // downcast an address to a reference
  referenceof : forall @(A : Type1),
    Address -> Ledger -> Option (Ref A) * Ledger

  // create a new immutable shared cell, accessible by everyone
  freeze : forall (A : Type), A -> Ledger -> Address * Ledger

  // read contents of an immutable cell
  peek : forall @(A : Type), Address -> Ledger -> Option A * Ledger
```

The above code listing shows the most important primitives (built-ins) available in Axi. `Address` is the data type of addresses, including accounts. `Ledger` is an abstract type that represents (the current state of) the ledger. The body of a transaction is a function of type `Ledger -> Ledger`.

Every function that interacts with the ledger takes (the current version of) the ledger as input and also returns (the new version of) the ledger as output. In general, the type of such a function is `A -> Ledger -> B * Ledger`, i.e. it takes a value of some type `A` and the ledger, and returns a value of some type `B` and the (updated) ledger.

The transaction body will, in general, be composed of many such functions. It will get the initial input of type `Ledger` from the transaction context and then pass it to the first of these functions, which will in turn pass it to the second one, and so on. Because `Ledger` is a resource type, the type system guarantees

that values of this type cannot be copied or deleted. This means that the ledger is acted upon and then passed further and further until the transaction body returns the final modified version of it.

An example function that needs the ledger is `whoami`, which consumes the old ledger and returns the account address that is the originator of the current transaction, together with a new ledger.

`new` is a function that takes an account address and a value of some type, together with the ledger, creates a new cell on the ledger that contains this value and belongs to that account, and returns the address of the created cell together with the new ledger.

`delete` is a function that takes a cell address, together with the ledger, and returns an optional value of some type `A`, together with the new ledger. If the transaction originator is not the owner of the cell, or the value stored in the cell is not of type `A`, then the operation fails and the result is `none`. Otherwise the operation succeeds, the cell is destroyed and the result is `some a`, where `a` is the value that was stored in this cell.

`share` is a function that takes a value of some type `A`, together with the ledger, creates a new mutable shared cell that contains this value, and returns the reference to the created cell together wtih the new ledger.

`freeze` is a function that takes a value (which must be data and not a resource), together with the ledger, creates a new immutable shared cell that contains this value, and returns the address of the created cell together with the new ledger.

`peek` is a function that takes an address, together with the ledger, reads the contants of an immutable shared cell under this address, and returns an optional value of some type `A` together with the new ledger. If the address does not point to an immutable shared cell, or the value stored in that cell is not of type `A`, then the operation fails and the result is `none`. Otherwise the operation succeeds and the result is `some a`, where `a` is the data stored in the cell.

`update` is a function that takes a reference to a shared cell, together with the ledger, and returns a value of some type `A` together with a callback that when called with the new value of the cell, returns the new ledger. Note that this callback has a resource type `A -> Ledger`, so the type system makes sure that the callback is eventually called once.

C.2 The resource lifecycle

A new resource type is created by a transaction that deploys a package of Axi code that contains the definition of this resource type. The only exception is Gas, the native coin used to pay transaction fees, which is built-in, so it already exists at the origin block. Because packages live on-chain forever and cannot be deleted, a resource type, once it comes to life, will also exist forever.

The life cycle of a particular resource, i.e. an element of some resource type, consists of three stages:

- **Creation**. At some point the resource is created according to the rules set by the package that contains the definition of the resource's type. After creation, the resource is placed on the ledger in a cell owned by somebody.
- **Transaction(s)**. The resource might participate in transactions, if its owner wishes so. The owner might transfer his ownership of the resource

freely, whereas programmatic interaction of the resource with other objects happens only according to the rules set by the defining package.

- **Destruction (optional)**. If the defining package permits it, the resource might at some point be destroyed. Examples of destructible resources are stablecoins that are backed by an off-chain reserve, where a coin gets destroyed when its off-chain collateral is removed from the treasury, and a wrapper for an asset bridged from another blockchain, which gets destroyed when the external asset ceases to be bridged.

The most important part of the lifecycle, in which most resources will spend almost all their life, is the phase when they can participate in transactions. An ordinary transaction (i.e. one which does not deploy a new package) consists of an arbitrary Axi program that transforms the state of the ledger, which we'll call the *transaction body*, and of some metadata, most importantly the *transaction originator*. This program, when executed, interacts with the ledger in three steps:

- **Withdraw**. First, resources are "withdrawn" from the ledger into the program. The ledger's ownership model ensures the safety of this step, for example by making sure that the program cannot withdraw resources that don't belong to the transaction originator.

- **Transform**. Second, the program transforms its resources by performing various computations, exchanging the resources for other resources and moving them around, etc. The safety of this step is guaranteed by the type system. The resources cannot get unlawfully destroyed or copied because the type system keeps an eye on them, nonsensical operations cannot be performed on them, because the program is well-typed, they cannot get lost in the jungle of pattern matching, because exhaustiveness is checked, and so on.

- **Deposit**. Third, the program "deposits" the transformed resources back onto the ledger. The safety of this step is partially ensured by the type system, which checks that all resources held by the program are deposited back. However, since resources can be freely transferred, it's up to the transaction author to make sure that they are transferred to the correct address and not to somebody else.

C.3 A longer example

Example 8. *Donut Shop*

```
module DonutShop where

// exporting the types would allow public access
// to constructors and record updates
// since the types are mentioned in the signatures
// the types will be public but opaque
export
  init
  buyDonut
  collectProfits
  getMyShopRef
```

```

// Donut - simple purchasable object, can be used at most once
record type Donut : Type? where
  serialNumber : Nat

// Donut shop - shared object
record type DonutShop : Type1 where
  thisShop : Address
  price : Nat
  balance : KhalaniCoin
  nextSerialNumber : Nat

record type DonutShopOwnershipToken : Type1 where
  myShopRef : Ref DonutShop

// Create a donut shop
init (ledger : Ledger)
: DonutShopOwnershipToken * Ledger
= let (shopRef, shopToLedger) = share ledger in
  let ownership : DonutShopOwnershipToken =
    record where
      myShopRef = shopRef in
  let shop : DonutShop = record where
    thisShop = addressof shopRef
    price = 1000
    balance = coinZero
    nextSerialNumber = 0 in
  (ownership, shopRef, shopToLedger shop)

// One can buy a donut for a set price
buyDonut (payment : KhalaniCoin) (shop : DonutShop)
: Option Donut * KhalaniCoin * DonutShop
= let (splitResult, change) = coinSplit shop.price payment in
  match splitResult with
  | none => (none, change, shop)
  | some paid =>
    let newBalance = coinMerge shop.balance paid in
    let newSerialNumber = shop.nextSerialNumber in
    let newShop : DonutShop =
      record shop where
        balance = newBalance
        nextSerialNumber = newSerialNumber + 1 in
    let donut : Donut = record where serialNumber = newSerialNumber in
    (some donut, change, newShop)

// Collects profits from shop (succeeds only for owner)
collectProfits (ownership : DonutShopOwnershipToken) (shop : DonutShop)
: KhalaniCoin * DonutShop
= if addressof ownership.myShop == shop.thisShop then
    let profits = shop.balance in
    let newShop = record shop where balance = coinZero in
    (profits, newShop, ledger)
  else
    (coinZero, shop, ledger)

// Reads shop reference from an ownership token
getMyShopRef (ownership : DonutShopOwnershipToken)
: DonutShop * DonutShopOwnershipToken
= (token.myShopRef, token)
```

```
from DonutShop import
  init

```

```
main (ledger : Ledger) : Ledger =
: Ledger -> Ledger
| ledger =>
  let (myAddress, ledger) = whoami ledger in
  let (ownershipToken, ledger) = init ledger in
  let (tokenAddress, initTokenCell) = freeze myAddress ledger in
  let ledger = initTokenCell ownershipToken in
  ledger
```

```
from DonutShop import
  DonutShop
  buyDonut

coinAddr
: Address
= 0x0664eebd187e0668e41e5d2236eaafe2d54dd9dbc76685e4ceac69c908c26446

shopAddr
: Address
= 0xf09785359712726892a2ec3b8b7cf460f0f6dd82e0bf5a7a8d2f2276035888cf

main
: Ledger -> Ledger
| ledger =>
  let (myAddress, ledger) = whoami ledger in
  let (optShopRef, ledger) = referenceof DonutShop shopAddr ledger in
  match optShopRef with
  | some shopRef =>
    let (optCoin, ledger) = delete KhalaniCoin coinAddr ledger in
    match optCoin with
    | some paymentCoin =>
      let (shop, returnShop) = update shopRef ledger in
      let (optDonut, changeCoin, shop) = buyDonut paymentCoin shop in
      let ledger = returnShop shop in
      let (newCoinAddr, initCoinCell) = new myAddress ledger in
      let ledger = initCoinCell changeCoin in
      match optDonut with
      | some donut =>
        let (newDonutAddr, initDonutCell) = new myAddress ledger in
        let ledger = initDonutCell donut in
        ledger
      | none => ledger
    | none => ledger
  | none => ledger
```

```
from DonutShop import
  DonutShopOwnershipToken
  collectProfits
  getMyShopRef

ownershipTokenAddr
: Address
= 0x22bf7f5182a7d22decc11f2a8fd94e2c6834cebd2ab2b239cab598987454bc1b

main
: Ledger -> Ledger
| ledger =>
  let (myAddress, ledger) = whoami ledger in
  let (optOwnershipToken, ledger) =
    delete DonutShopOwnershipToken ownershipTokenAddr ledger in
  match optOwnershipToken with
  | some ownershipToken =>
```

```
    let (shopRef, ownershipToken) = getMyShopRef ownershipToken in
    let (shop, returnShop) = update shopRef ledger in
    let (profitsCoin, shop) = collectProfits ownershipToken shop in
    let ledger = returnShop shop in
    let (newCoinAddr, initCoinCell) = new myAddress ledger in
    let ledger = initCoinCell profitsCoin in
    let (newTokenAddr, initTokenCell) = new myAddress ledger in
    let ledger = initTokenCell ownershipToken in
    ledger
  | none => ledger
```